Which Machine Learning Models Are Actually Used in Industry?

Student Report

Introduction

Machine learning is now used in many companies and organizations. But the most famous models are not always the most used models in real work. In this report I review industry surveys and technical reports to answer one main question. Which machine learning models are most widely used in real industry today?

Most Common Models in Industry Today

The evidence shows that classical machine learning is still very important in industry. Many business problems use tabular data. This means the data is in rows and columns such as customer age, price, number of orders, payment history, or medical values. For this kind of data tree based models are very common. Popular examples are XGBoost, LightGBM, CatBoost, Random Forest, and Decision Trees.

These models are used a lot because they work very well on tabular data. They usually need less data than deep learning models. They are also faster to train and easier to use in many business systems. In many cases they give strong results without very heavy computing.

Linear Regression and Logistic Regression are also still widely used. They are simple, fast, and easy to explain. This is very important in areas like banking, insurance, healthcare, and public services. In these areas companies often need a model that managers, clients, or regulators can understand.

Deep learning models are also important, but mostly in tasks with unstructured data. For images companies use CNN based models and transfer learning. For text and language they use RNNs in older systems and Transformers in newer systems. In speech, vision, and language tasks deep learning is often the best choice.

Large Language Models are growing very fast. They are used in chatbots, search, document analysis, coding help, customer support, and knowledge systems. But this does not mean they have replaced all older models. In many normal business prediction tasks the older models are still more practical.

Why Classical Models Are Still So Common

I think there are four main reasons. First, most companies still work with structured tabular data. Second, classical models are cheaper and easier to train. Third, they are often easier to explain and monitor. Fourth, many business teams do not need the extra complexity of deep learning for every task.

This is why the most advanced model is not always the most useful model. In real industry people care about accuracy, speed, cost, stability, and explainability at the same time.

Prediction for the Next 5 to 10 Years

In the next five to ten years I believe classical models will remain very important. Linear models and tree based models will still be strong in finance, risk scoring, pricing, forecasting, operations, and many tabular prediction tasks. These areas need stable and explainable systems.

At the same time deep learning and LLM based systems will gain more share. The biggest change will happen in domains that depend on text, images, audio, video, and human interaction. Customer service, document processing, software development, search, education tools, and digital assistants will likely use more Transformer and LLM based systems.

So the future will probably not be one model replacing all others. Instead we will see a mixed world. Classical models will stay important for structured business data. Deep learning and LLMs will grow strongly in language and multimodal tasks. Many companies will use hybrid systems that combine both.

Conclusion

Based on current reports and industry behavior, the most widely used family of models in real business settings is still classical machine learning for tabular data. Tree based models are especially strong in practice. Linear and Logistic Regression are also still very common. Deep learning is dominant in image, speech, and language tasks. LLMs are growing fast, but they are adding to the industry model mix rather than fully replacing older models today.

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